

Evidence-Traceable Knowledge Graph Construction from Marine Pump Technical Documents via Automated Evidence-Quality Gating

Xiaokun Zhang¹, Dongliang Fu^{1,*}, Rui Wang², and Erwu Liu²

¹Shanghai Marine Equipment Research Institute (SMERI), China State Shipbuilding Corporation (CSSC), Shanghai, China

²Tongji University, Shanghai, China

*Corresponding author: Dongliang Fu; e-mail: fdl@163.com

Abstract—Measurements from marine engine-room pumps become more interpretable and auditable when linked to documented causes, inspections, and maintenance actions. Large language models can extract such relations from manuals, but their outputs may contain unlocatable evidence, reversed relations, misaligned table cells, or duplicated support copied across documents. This paper presents an automated evidence-quality gating method for constructing a page-traceable and auditable knowledge graph. Each normalized triple is stored separately from every page-level evidence record that supports it. Admission requires valid graph structure and direction, recoverable page or table evidence, relation support, complete provenance, build-corpus membership, and non-inference. A full audit graph retains all candidates and gate outcomes; evidence-qualified records with normalized Chinese terminology form tiered application graphs. From 15 documents and 1,934 physical pages, the pipeline produced 8,003 audit records, of which 1,698 passed all evidence gates. The standard application layer contains 1,326 evidence records and 1,281 unique triples, covering 78.09% of the evidence-qualified set. On 20 fixed pages, complete gating retained 148 records with both endpoints recoverable inside the located evidence, compared with 223 of 243 records retained by simple filtering of the same candidates. All 1,698 evidence-qualified records passed re-localization and provenance checks; all 13,729 controlled rule violations triggered a gate. Four held-out documents from one source family yielded 81 records meeting the non-corpus evidence gates. Eightfold replication within one source family did not change the family-capped support score. The method enables evidence-grounded graph construction without manual approval of every triple at admission time. Qualification denotes compliance with declared evidence rules, not expert-confirmed factual or diagnostic accuracy.

Index Terms—marine pump systems, knowledge graph, evidence-quality gating, page-level provenance, terminology normalization, condition-monitoring knowledge

I. INTRODUCTION

Pumps, motors, pipes, and valves in a ship's engine room support cooling, ballast transfer, bilge drainage, and fuel delivery. Condition-monitoring systems observe insufficient inlet pressure, reduced flow, elevated vibration, high temperature, and abnormal current [?], [?]. These measurements seldom identify a fault alone; engineers consult classification guidance, manuals, and maintenance tables to link them to possible causes and actions.

Such knowledge is evidence-dependent. A relation may appear in prose, a troubleshooting table, or a maintenance procedure; tables may contain merged rows and inherited headers. Retaining only a deduplicated head–relation–tail triple loses the exact supporting passage. Moreover, manuals from one source often repeat content, so document count overstates independent support.

Knowledge graphs organize equipment, faults, symptoms, inspections, and maintenance actions [?], [?]. LLMs reduce extraction effort, but schema alignment, canonicalization, and output stability still require explicit controls [?]. Industrial fault graphs mainly address extraction, fusion, reasoning, or diagnosis [?], [?], [?]. A preceding issue remains: what document-evidence conditions should an automatically extracted record satisfy before entering an application graph?

We treat each generated triple as a candidate, not a fact. It must point to recoverable evidence and pass frozen executable checks for field completeness, structure, direction, localization, provenance, and corpus boundaries. Passing denotes compliance with these admission rules, not expert confirmation of objective truth.

The study asks whether the pipeline produces schema-compliant, page-localized, provenance-complete records (RQ1); how evidence and terminology gates affect record retention and graph size (RQ2); whether frozen rules transfer to held-out sources (RQ3); whether source-family capping resists within-family duplication (RQ4); and how terminology tiers affect predefined fault-path coverage and density (RQ5).

The contributions are threefold. First, normalized triples are separated from page-level evidence records, preserving multiple supports after semantic deduplication. Second, executable gates jointly determine qualification, quarantine, or rejection and retain a complete audit trail. Third, tiered Chinese application graphs are derived from the audit graph, while source-family capping limits repeated documents assigned to one family from appearing as independent corroboration. We evaluate these mechanisms by complete-candidate re-audit, controlled fault injection, terminology sensitivity, source hold-out, and replication stress tests.

II. RELATED WORK

A. LLM-Based Knowledge Graph Construction

Knowledge graph construction combines extraction, schema alignment, canonicalization, and fusion [?], [?]. Extract-Define-Canonicalize separates open extraction, schema definition, and post-hoc canonicalization, showing why LLM construction is not a single prompting step [?]. Industrial graphs support process monitoring, rotating machinery, and equipment diagnosis [?], [?], [?]. Our focus is earlier: the evidence requirements that generated records must meet when manual review of every triple is not a prerequisite.

B. Statement-Level Provenance and Evidence Support

W3C PROV-O represents derivation through entities, activities, and agents [?]; named or contextualized graphs can retain statement-level provenance [?]. RDF 1.2 triple terms and reifiers describe a statement without directly asserting it [?], while nanopublications separate an assertion from provenance and publication information [?]. ProVe verifies textual support for graph triples [?]; SciGraph-LLM jointly extracts atomic claims, source spans, and provenance [?].

Marine documents add physical-page and table localization, directed relations, Chinese terminology, and repeated content within source families. Rather than proposing a provenance standard, we turn established representation principles into an executable admission workflow.

III. METHOD

A. Corpus Partition and Page Parsing

MP numbers are corpus registry IDs, not a continuous build-set sequence. The build set comprises MP001–MP007 and MP015–MP022: 15 documents and 1,934 physical pages. MP008 supported parser, prompt, and rule debugging; its two retained pilot candidates do not establish statistical parameter calibration. MP009 was screened before the split and provides only a non-blind transfer case. MP010–MP013 were first processed after method and graph freezing, without feedback into parameter selection. MP008–MP013 remain outside the primary graph. MP014 was excluded because it is an offshore electric-submersible-pump signal dataset rather than a marine technical document.

Deterministic rules removed blank, cover/title, contents, index, and exact-duplicate pages, leaving 1,889 extraction pages. Each page object retained text, table cells, bounding boxes, physical page, URL, publisher, source family, and document/page hashes. Candidate generation used the Alibaba Cloud Model Studio alias qwen3.7-max through the Dash-Scope API at temperature zero on July 26–27, 2026; requests and responses were cached. The API records contain the alias rather than a dated model snapshot, so the historical backend snapshot cannot be established retrospectively.

The schema contains 13 node and 15 relation types. Types, directions, the whitelist, source families, and thresholds were frozen before holdout processing. Build-set results are not

independent of earlier iterative rule design. Terminology experiments and the revision’s parameter perturbations retain frozen evidence decisions and select no replacement settings.

B. Normalized Triples and Page-Level Evidence Records

A normalized triple is denoted by

$$c_i = (h_i, r_i, t_i), \quad (1)$$

where h_i , r_i , and t_i are the head entity, relation, and tail entity. Every occurrence of document support creates a separate page-level evidence record:

$$a_i = \langle c_i, e_i, p_i, g_i \rangle, \quad (2)$$

Page-level evidence (e_i) is the supporting passage or table cells; page-level provenance (p_i) records document identity, physical page, URL, publisher, source family, and hashes. The component g_i stores evidence type, support mode, score, and gate outcomes. A triple may link to multiple evidence records; semantic deduplication never overwrites page-level support.

For example, MP016 p. 3 states that insufficient inlet pressure causes pump cavitation. The model proposes *insufficient inlet pressure—causes—cavitation*. The system validates its types and direction, relocates both endpoints and the relation phrase, and stores page, URL, hashes, and source family. A repeated sentence in another manual adds evidence but not another contribution from that family.

C. Automated Evidence-Quality Gating

The gates comprise four groups.

Structure: Endpoint types, relation, and direction must comply with the whitelist and schema.

Evidence: E1 requires both entities in one continuous page span. E2 requires relevant cells in one row or validated merged-row group, resolvable cell locations, column names, and an affirmative layout-validation flag. Re-localizing cell coordinates alone does not satisfy all E2 admission conditions. Cross-sentence E3 records remain auditable but cannot enter an application graph.

Relation support: Explicit rules require an unambiguous phrase or table role. Otherwise, the extraction model receives support- and rejection-oriented prompts. Both must support the relation at confidence ≥ 0.9 , with relocatable quotations.

Admission boundary: The governance score $q_i = \alpha_i w_E(l_i) w_R(\eta_i)$ combines model-reported confidence α_i with frozen weights for evidence level and relation-support status. The weights for E1/E2/E3 are 1.00/0.95/0.75, and those for supported/undetermined/unsupported relations are 1.00/0.75/0. A record must belong to the build set, be non-inferred, contain complete URL/family/hash provenance, and have $q_i \geq 0.8$. This heuristic supports filtering; it is not a factual probability.

The weights express a heuristic ordering, not calibrated probabilities: E1 and supported relations provide the unit reference, E2 receives a small alignment penalty, and reconstructed or unresolved support is discounted. No expert calibration or optimization search is documented. The 0.80 threshold

requires confidence of at least 0.80 for E1 and approximately 0.8421 for E2, after mandatory evidence checks. Section IV-C examines sensitivity.

The combined gate is

$$H(a_i) = I_{str}I_{dir}I_{loc}I_{sup}I_{prov}I_{build}I_{noninf}I_{q \geq 0.8}. \quad (3)$$

All indicators must equal one for evidence qualification. E3 and unresolved relations cannot qualify regardless of their weights. Recoverable cases enter quarantine if their final or applicable structural score reaches the separate 0.60 candidate floor; fatal failures or lower scores cause rejection. This floor differs from the 0.80 admission threshold.

D. Full Audit Graph and Tiered Chinese Application Graphs

The *full audit graph* retains every candidate, gate outcome, and failure reason. The *Chinese application graph* selects **evidence-qualified**, non-inferred records for retrieval and visualization, then assigns standard, conservative, or strict layers from both endpoints' terminology status. Stable IDs link each application record to the audit graph and physical page (Fig. 1).

Normalization never rewrites verbatim evidence. Frozen dictionary and earlier two-prompt terms are retained. Other Chinese names require completeness, no same-type naming conflict, protected-string preservation, and the layer threshold. Standard uses 0.90; conservative uses 0.90 for recurring endpoints and 0.95 for singletons; strict retains only the original frozen status. Protected strings include models, units, API, ISO, and NPSH. These checks indicate local consistency, not independent or expert confirmation.

E. Source-Family Capping

Manuals from one publisher may share content. For triple c , only the strongest evidence score in each publisher- or document-defined family is retained:

$$s_f(c) = \max_{a_i \in A_f(c)} q_i, \quad S_{fam}^{(B)}(c) = \frac{1}{B} \sum_{j=1}^{\min(B, |F_c|)} s_{(j)}(c). \quad (4)$$

Here F_c is the set of families supporting c , $s_{(j)}$ is the j th-largest family score, and $B = 2$ is the frozen engineering coverage target; missing family slots contribute zero. The index is audit and ranking metadata and does not change gate decisions or application-graph membership. Duplicates within one family therefore cannot increase its contribution. The family label is an engineering proxy for dependence, not proof of independence.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Evaluation covered full-corpus gating, re-audit, fault injection, fixed-page baselines, terminology layers, parameter sensitivity, source holdout, replication, and fault paths. The revision analyses reused frozen candidates and cached responses, made no API calls, and verified that input hashes were unchanged. No human factual labels were available.

TABLE I
PAGE LOCALIZATION AND PROVENANCE RE-AUDIT

Decision	Records	Evidence	Endpoints	Subchecks
Evidence-qualified	1698	100.00%	100.00%	100.00%
Quarantined	881	100.00%	100.00%	62.54%
Rejected	5424	72.77%	48.80%	31.47%

TABLE II
CONTROLLED FAULT-INJECTION RESULTS

Fault group	Corrupted fields	Cases	Detection
Identity/provenance	hash, URL, family	6792	100.00%
Page/evidence	page, span, endpoints	4943	100.00%
Structural type	relation-tail type	1698	100.00%
Table structure	cell ID, bounding box	296	100.00%
Total	10 fault types	13729	100.00%

B. Complete Candidate Re-Audit and Controlled Fault Injection

The 8,003 records referenced 922 physical pages. Without reading original decision labels, re-audit reloaded page objects and checked document-page mappings, hashes, URLs, publishers, families, evidence and endpoint spans, and E2 table locations.

Table I shows that all 1,698 **evidence-qualified** records passed localization, provenance, and type checks. All 881 quarantined records retained locatable evidence and endpoints; 330 failed the listed subchecks because they used E3 reconstruction. Of 5,424 rejected records, 3,947 retained locatable evidence, 2,647 both endpoints, and 1,707 all listed subchecks. Of those 1,707, 1,451 still failed original E2 admission requirements: 1,449 lacked the affirmative layout flag and 290 lacked column names, with overlap. A further 253 failed relation support and three fell below the 0.60 candidate floor. The re-audit checks recoverability, but does not replace these recorded admission requirements.

Table V assigns each rejection to its first failed category in the stated order, not to a gate's independent causal effect. Flags overlap: 4,632 records have evidence failures and 2,777 have rule-based relation-support failures.

Fault injection corrupted one declared rule at a time in the 1,698 **evidence-qualified** records. Ten fault types covered identity/provenance, page/evidence location, endpoint types, and tables. Every valid mutation triggered a gate (Table II); this tests encoded-rule response, not general semantic-error detection.

C. Full-Corpus Graph, Terminology Layers, and Gate Contributions

From 8,071 raw candidates, normalization and deduplication produced 8,003 audit records: 1,698 **evidence-qualified**, 881 quarantined, and 5,424 rejected. The **evidence-qualified records** represent 1,650 unique triples and 2,613 source-language entities.

By relation-support mode, explicit phrases supported 784 records and table roles supported 98; the other 816 used

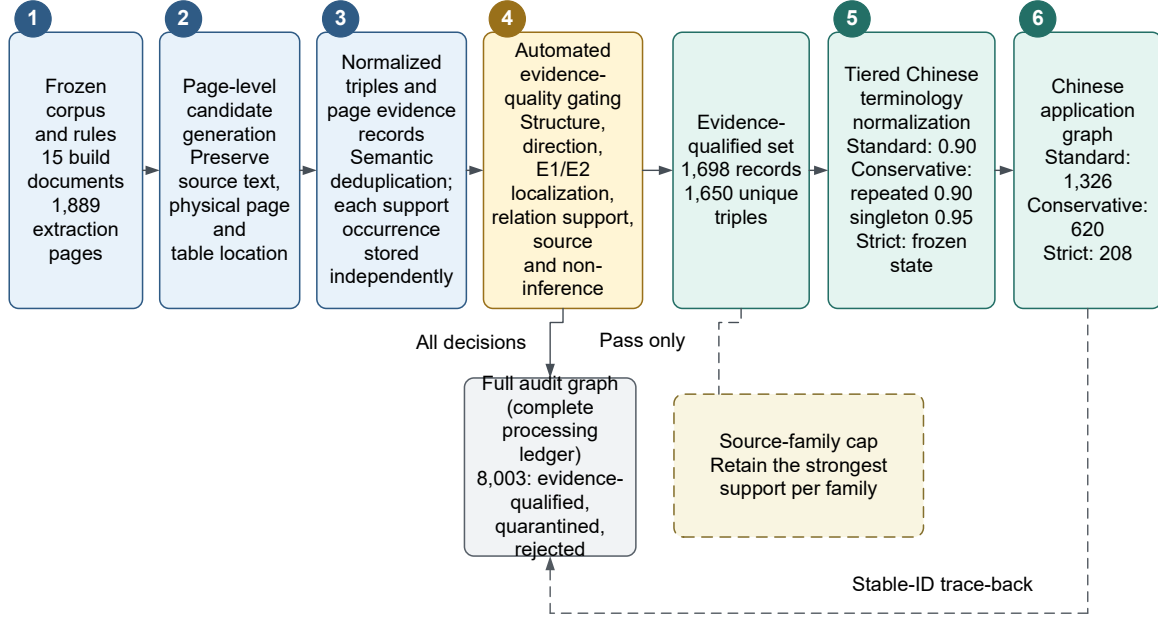


Fig. 1. Workflow for automated evidence-quality gating, the complete processing ledger, and tiered Chinese application graphs.

TABLE III
EFFECT OF CHINESE TERMINOLOGY LAYERS ON APPLICATION-GRAPH SIZE AND FAULT-PATH DENSITY

Layer	Terminology rule	Evidence records	Unique triples	Chinese entities	Path coverage	Mean answers	Mean evidence
Strict	Frozen dictionary and two-prompt status	208	203	281	34/40	5.075	6.150
Conservative	Recurring 0.90; singleton 0.95	620	590	870	34/40	5.575	6.800
Standard	Uniform threshold of 0.90	1326	1281	2022	34/40	6.400	7.700

TABLE IV
SIMPLE BASELINES AND GATING ON THE SAME 20 PAGES

Configuration	Records	Evidence	Both endpoints
Direct LLM	162	45.06%	40.12%
Direct LLM + basic	115	61.74%	56.52%
Shared: no gate	367	87.74%	79.56%
Shared: basic	243	97.94%	91.77%
Shared: full	148	100.00%	100.00%

TABLE V
PRIMARY FAILURE CATEGORIES AMONG REJECTED RECORDS

Attribution order and category	Records	Share
1. Schema/type incompatibility	1292	23.82%
2. Evidence admission failure	3868	71.31%
3. Relation support failure	253	4.66%
4. Below 0.60 candidate floor	11	0.20%
Total	5424	100.00%

same-model two-prompt consistency and were marked separately because it is not independent validation. By evidence-localization mode, the evidence-qualified set comprised 1,550 E1 and 148 E2 records.

Before score filtering, 1,704 records met the remaining hard gates; 1,698 passed the 0.80 threshold. This six-record reduction concerns admission, whereas Table V concerns the separate rejected-record population.

Terminology processing kept all evidence decisions fixed (Table III). The standard layer covers 78.09%, with 6.38 times as many records as strict. Its 372 exclusions involve 286 unresolved endpoint groups: 205 below 0.90 confidence, 75 with missing or conflicting Chinese names, and six failing protected-string preservation. Endpoints recur, and either endpoint can block a record.

MP003, MP005, and MP022 contribute 149, 74, and 39 exclusions, with within-document rates of 29.04%, 27.82%, and 40.21%. Maintenance, inspection, and prevention relations contribute 118, 84, and 58 exclusions, with rates of 25.99%, 22.40%, and 34.32%. The longer endpoint has a median of nine words among excluded records, versus seven among included records. Lowering only the terminology threshold to 0.88 or 0.85 would admit 1,329 or 1,491 records. These uneven losses expose a coverage cost; without semantic labels, they neither establish the additional names' correctness nor rule out overfitting.

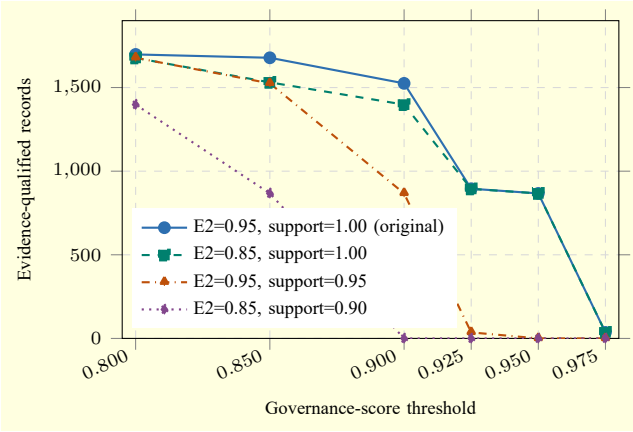


Fig. 2. Joint weight and threshold sensitivity with all hard gates fixed.

Both expanded layers passed 36 release checks covering integrity, IDs, evidence links, splits, evidence types, provenance, schema, terminology, and verbatim preservation. A non-blocking warning concerned 40 rejected records lacking verbatim text. Evidence and terminology thresholds govern separate decisions.

A joint sweep on 1,704 hard-gate survivors crossed four E2 weights (0.85, 0.90, 0.95, 1.00), three supported-relation weights (0.90, 0.95, 1.00), and seven thresholds (0.75, 0.80, 0.85, 0.90, 0.925, 0.95, 0.975): 84 combinations. At threshold 0.80, counts range from 1,398 to 1,698. With support weight 1.00, E2 weights of 0.90–1.00 retain 1,697–1,698 records. At E2 weight 0.95, support weights of 0.95 and 0.90 retain 1,678 and 1,525. E3 and undetermined weights cannot affect admission under their hard vetoes.

Figure 2 shows the joint sensitivity. Original weights retain 1,698 records at 0.80 but only 36 at 0.975. Higher thresholds expose clustered confidence values. The post-submission sweep describes sensitivity, not optimality or a historical parameter-selection procedure.

D. Simple Baselines and Prompt-Rule Ablation

Table IV uses the original 20 evidence-rich build pages. Basic filtering requires nonempty triple and type fields, matches both endpoints anywhere on the page after case/whitespace normalization, and deduplicates within-page triples. It uses no evidence-span, table-alignment, relation-support, or score constraint. Direct LLM outputs keep their native schema. All rows use the same page/cell re-localizer; “both endpoints” requires both entities inside the located evidence.

The lower three rows fix the full prompt’s 367 normalized candidates. Basic filtering retains 243 records, with 223 (91.77%) passing endpoint localization; full gating retains 148, all passing. Basic filtering retains only 98 of those 148: the remaining 50 are E2 records recoverable from cells but missed by flat-page matching. Thus basic filtering can retain poorly grounded records and omit recoverable table evidence. Parser-supplied provenance is complete in every row, not a comparative gain.

Historical B0–B3 are prompt-rule ablations (unconstrained, schema, evidence, and direction/applicability), yield-

ing 0/50/31/32 evidence-qualified records versus 148 for the full prompt. Unmatched candidate/token limits prevent claims of extraction superiority. The shared-candidate comparison isolates post-processing at fixed extraction cost; direct-LLM rows provide a workflow reference. Neither measures factual precision, recall, or F1.

E. Source Holdout, Within-Family Replication, and Fault Paths

After freezing the method and primary graph, MP010–MP013 underwent first extraction in isolation. Their 103 physical pages (94 after exclusions) yielded 287 audit records. Of these, 81 met the non-corpus evidence gates with complete page localization and provenance but retained held-out status under the build-boundary rule. They did not feed back into tuning. All belong to the MAIB family, so this test covers structural and evidence-rule behavior on unseen documents, not source diversity or external factual accuracy.

Three triples occurred identically in two documents from one family. Replicating evidence one, two, four, and eight times left the mean capped index at 0.461058 (maximum change zero). Direct document accumulation raised the mean from 0.461877 to 0.922116 and the number scoring ≥ 0.8 from three to 1,650. Thus within-family duplication cannot increase capped support; no natural identical cross-family triple was available to test cross-family discrimination.

Every terminology layer covered 34/40 predefined fault paths (85%); the four groups remained 7/10, 10/10, 7/10, and 10/10. From strict to standard, mean answers rose from 5.075 to 6.400, page evidence from 6.150 to 7.700, documents from 2.700 to 3.075, and families from 2.350 to 2.600. Expansion increased density, not path types; these are not question-answering or diagnostic accuracy.

V. DISCUSSION

A. Meaning of Qualification and the Human-Review Boundary

Re-localization and controlled fault injection show that evidence qualification is a repeatable executable state. They do not establish factual correctness.

Terminology relaxation expands the application graph from 208 to 1,326 records without changing evidence gates. Stricter naming policies have not been shown to improve factual accuracy or diagnostic safety.

Reusable rules remove triple-by-triple expert approval as an admission prerequisite, but cannot make manuals infallible. Human judgment defines the rules; conflicts, safety-critical statements, and out-of-scope conditions still warrant targeted review.

B. Relation-Support Modes and Measurement Knowledge

Explicit phrases or table roles support 882 records; 816 use two checks by the extraction model. Their errors may be correlated, so the modes are exposed rather than treated as equally independent. Safety-critical retrieval can prioritize

explicit-rule records or target model-consistency records for review.

This work addresses the knowledge-data layer of measurement and diagnosis, not a new sensor or classifier. The graph links pressure, flow, vibration, temperature, and current observations to documented causes, inspections, and maintenance, with a page basis for every relation. It can support evidence retrieval, diagnostic assistance, and later GraphRAG systems; no sensor time series or fault-recognition accuracy is reported.

C. Limitations

The study evaluates evidence auditability rather than state-of-the-art extraction accuracy. No expert factual labels were available for precision, recall, or F1. Shared-model relation checks and terminology suggestions may have correlated errors. Fault injection tests encoded violations. The direct-extraction runs are not budget matched, whereas the shared-candidate control fixes extraction outputs and cost. Heuristic weights remain uncalibrated; the post-submission sweep measures sensitivity rather than optimality. Terminology exclusions vary by source and relation, so overfitting cannot be ruled out. Source-family testing lacks natural identical cross-family triples.

VI. CONCLUSION

Automated evidence-quality gating separates normalized triples from page support and preserves every admission

decision. Fifteen marine pump documents yielded 8,003 audit records, 1,698 evidence-qualified records, and a standard application layer of 1,326 records. Compared with basic filtering of shared candidates, full gating raises the endpoint-recovery rate and retains table evidence missed by page-wide matching. Rejection attribution, terminology analysis, and parameter sweeps clarify the coverage costs. The graph supplies traceable knowledge for measurement interpretation; factual and diagnostic accuracy remain unevaluated.

ACKNOWLEDGMENT AND DECLARATIONS

Data availability: Code, configurations, scripts, and releasable derived results will be released upon acceptance. Copyrighted PDFs will not be redistributed; source URLs are retained with the evidence records. **Funding:** This research received no external funding. **Conflict of interest:** The authors declare no conflict of interest. **Ethics:** No human participants, animals, or personal data were involved. **Generative AI:** The Alibaba Cloud Model Studio alias qwen3.7-max generated candidates, performed same-model relation checks, and suggested initial Chinese terms through the DashScope API on July 26–27, 2026. The API returned the alias rather than a dated snapshot; prompt versions, request identifiers, and responses were retained. Terminology expansion used local records only. Generative AI assisted English translation and language editing. The authors reviewed the manuscript and take responsibility for its content.